

# Rigid Analytic Geometry Tutorial

So Murata

SoSe 26, University of Bonn

## 1 Week 2

### 1.1 Motivation and Obstacles

The motivation for the course is that, given a non-archimedean field  $k$ , how should we define analytic varieties and presheaves of algebraic functions over  $k$ .

**Definition 1.1** (Classical Rigid Analytic Geometry). *Intuition, over  $\overline{B(0,1)}^n \subseteq K^n$ , the analytic functions should be of the form of convergent power series.*

$$f = \sum_{\alpha \in \mathbb{N}^n} a_{\alpha} x^{\alpha}.$$

*This is a local description of the functions. We denote the set of such functions*

$$\mathbb{T}_n(K) := K\langle x_1, x_n \rangle,$$

*which we call the Tate algebra.*

**Remark 1.1.** *There is a one-to-one correspondence between*

$$\mathrm{Sp} \, \mathbb{T}_n(K) \longleftrightarrow \overline{B_{\overline{K}}(0,1)}^n / \mathrm{Gal}(\overline{K}/K).$$

We will denote  $X = \mathrm{Sp} \, \mathbb{T}_n(K)$ . Furthermore, we can define admissible opens  $U \subseteq X$ , analytic functions on  $U$ , and the presheaf of analytic functions  $\mathcal{O}_X$ . Over a wide range of admissible open coverings,  $\mathcal{O}_X$  satisfies the sheaf requirement.

$$0 \rightarrow \mathcal{O}(X) \hookrightarrow \prod_i \mathcal{O}(U_i) \rightrightarrows \prod_{i,j} \mathcal{O}(U_i \cap U_j).$$

We only allow certain open coverings called admissible covers. We introduce and work on a Grothendieck topology.

**Remark 1.2.** *There's a non-trivial sheaf of abelian groups  $\mathcal{F}$  on  $X$  with  $G$ -topology such that  $\forall x \in X, \mathbb{F}_x = 0$ .*

To overcome the obstacle above, we will introduce extra points, namely van der Put points. We glue  $\mathrm{Sp} \, \mathbb{T}_n(K)/I$  along opens to get rigid  $K$ -spaces.

## 1.2 Raynaud's Formal Schemes

Suppose  $X$  is a  $\mathbb{Q}_p$ -scheme, we want to consider the reduction mod  $p$ . We find a  $\mathbb{Z}_p$ -scheme  $X'$  such that

$$X' \times_{\mathbb{Z}_p} \mathbb{Q}_p \cong X.$$

Note that we can't guarantee that  $X'$  is unique or even it exists. Take  $R$  to be the valuation ring of  $K$  for example  $K = \mathbb{Q}_p, R = \mathbb{Z}_p$ , we want to find an object the structure over  $R$ , such that its generic fiber is  $X$ . Say  $X = \mathrm{Sp} \mathbb{T}_n(K) = \mathrm{Sp} K\langle x_1, \dots, x_n \rangle$ , we can take

$$Y = \mathrm{Spf} R\langle x_1, \dots, x_n \rangle,$$

We also set,

$$Y^{\mathrm{rid}} = \mathrm{Sp}(R\langle x_1, \dots, x_n \rangle \hat{\otimes}_R K).$$

If  $R$  is separated  $\mathfrak{a}$ -adically topologized, that is for any subset  $I \subseteq R$ ,  $U$  is open if and only if there is  $n \in \mathbb{N}$  such that  $\mathfrak{a}^n \subseteq I$ , and  $\bigcup_{n \geq 1} \mathfrak{a}^n = 0$ . Then we have,

$$R\langle x_1, \dots, x_n \rangle = \varprojlim_{m \geq 1} R/\mathfrak{a}^m[x_1, \dots, x_n].$$

We have another construction of rigid spaces, namely Berkovich's theory.

## 1.3 Huber's adic spaces

Let  $k$  be algebraically closed and  $K/k$  is a finitely generated field extension of transcendence degree 1. Then there is a non-singular projective curve over  $k$  with function field  $K$ .

**Construction** Take  $C_K$  to be the set of discrete valuation rings in  $K/k$ . Take closed subsets to be the whole space and finite sets. On each open  $U$ , define

$$\mathcal{O}(U) = \bigcap_{p \in U} R_p,$$

then  $(C_K, \mathcal{O})$  is isomorphic to a non-singular curve with function field  $K$ .

Given a topological ring  $A$  and its subring  $A^\triangleleft$  or sometimes denoted  $A^+$  (integrally closed, open, power bounded). Huber defines,

$$\mathrm{Spa}(A, A^\triangleleft) = \{\text{continuous valuations } v : A \rightarrow \Gamma \cup \{0\} \text{ such that } \forall a \in A^\triangleleft, v(a) \leq 1\} / \sim.$$

**Remark 1.3.** *There is a fully-faithful functor*

$$r_K : \mathrm{Rig}(K) \rightarrow \mathrm{Adic} / \mathrm{Spa}(K, K^\circ),$$

*from the category of rigid  $K$ -spaces, to the category of adic spaces over  $\mathrm{Spa}(K, K^\circ)$ .*

**Remark 1.4.** *There is a fully-faithful functor*

$$t : \mathrm{FSch}^{\mathrm{ln}} \rightarrow \mathrm{Adic} / \mathrm{Spa}(K, K^\circ),$$

*from the category of locally Noetherian formal schemes, to the category of adic spaces over  $\mathrm{Spa}(K, K^\circ)$ .*

**Remark 1.5.** *If  $X$  is a qcqs rigid  $K$ -space, the set of van der Put points of  $X$  is one-to-one correspondent to points of  $r_K(X)$ .*

The problem session goes parallel to the lecture, meaning the lecture will focus on the modern treatment while the tutorial will focus on the classical one. Remark 1.5 will be the meeting point of these two.

## 2 Week1

**Problem 2** Let  $f, g \in \mathbb{T}_m$ . Set

$$f = \sum_{\alpha} a_{\alpha} x^{\alpha}, g = \sum_{\beta} b_{\beta} x^{\beta}.$$

Clearly,

$$\begin{aligned} \|f + g\| &= \max_{\alpha} |a_{\alpha} + b_{\alpha}|, \\ &\leq \max_{\alpha} \{|a_{\alpha}|, |b_{\alpha}|\}, \\ &= \max\{\|f\|, \|g\|\}. \end{aligned}$$

For  $k \in K$ , we have,

$$\|k \cdot f\| = \max_{\alpha} |ka_{\alpha}| = |k| \max_{\alpha} |a_{\alpha}|.$$

For the multiplicity, assume  $\|f\|, \|g\| = 1$ . This means there are  $\alpha_0, \beta_0$  such that

$$a_{\alpha_0} = 1, b_{\beta_0} = 1.$$

Recall that  $K^{\circ}$  is a local with maximal ideal  $K^{\circ\circ}$  so set  $k = K^{\circ}/K^{\circ\circ}$ . Let  $\bar{f}, \bar{g}$  be reductions of some  $f, g$  in  $k[x_1, \dots, x_n]$ . Then  $\bar{f}, \bar{g}$  are non-zero so  $\bar{f}\bar{g}$  is non-zero. This shows that some coefficients of  $fg$  is in  $K^{\circ} \setminus K^{\circ\circ}$ . Thus  $\|fg\| \geq 1$ .

We have,

$$\|fg\| \leq \max_{\gamma} \max_{\alpha+\beta=\gamma} |a_{\alpha} b_{\beta}| \leq \|f\| \|g\|.$$

**Definition 2.1** (Topological and Norm-free definition of boundedness, Power boundedness). *Let  $A$  be a topological ring and a subset  $X$  of  $A$  is called bounded if for any neighbourhood  $U$  of 0, there is a open neighbourhood  $V$  of 0 such that*

$$V \cdot X \subseteq U.$$

*An element  $a \in A$  is called power-bounded if  $\{a^n \mid n \geq 1\}$  is bounded.*

**Example 2.1.** *Consider  $A = k[x]/(x^2)$ .*

**Remark 2.1.** *If we consider the quotient topology on  $A/I$  for  $I$  open, then  $A/I$  will be equipped with discrete topology. This is because  $a + I$  is open thus the reduction  $\{\bar{a}\}$  is open.*

If  $A$  is a  $K$ -Banach algebra and the topological structure of  $K$  is not discrete by definition. An open ideal  $I$  in  $A$  will make  $K$ -Banach algebra  $A/I$  discrete, in particular,  $K \subseteq A/I$  is discrete thus  $A/I$  is trivial.

Let  $R = K^\circ$  be the valuation ring of  $K$ . We then define,

$$R\langle x_1, \dots, x_n \rangle = \left\{ f = \sum_{\alpha \in \mathbb{N}^m} a_\alpha x^\alpha \mid |a_\alpha| \rightarrow 0 \right\}.$$

Define the topology on  $R\langle x_1, \dots, x_n \rangle$  as the subset topology of  $\mathbb{T}_m$ . Then this admits a non-trivial open proper ideal. We take a pseudo-uniformizer  $\varpi \in R^{\circ\circ} = K^{\circ\circ}$  (i.e.  $|\varpi| < 1$ ), then  $(\varpi)$  is an open ideal of  $A$ . Moreover,

$$\{(\varpi^n)\}_{n \in \mathbb{N}} = \left\{ \sum_{\alpha \in \mathbb{N}^m} a_\alpha x^\alpha \mid |a_\alpha| \rightarrow 0, |a_\alpha| \leq |\varpi^n| \right\}.$$

is a basis of neighborhoods of 0.

What is the completion of  $R[x]$  with respect to  $(\varpi)$ ?

**Definition 2.2.** *The completion of  $R[x]$  with respect to  $(\varpi)$  is defined as*

$$\varprojlim_n R[x]/(\varpi)^n = R\langle x \rangle, \varprojlim_n R[x]/x^n = R[[x]], \varprojlim_n R[x]/(x, \varpi)^n = R[[x]].$$

Question is

$$R\langle x_1, \dots, x_n \rangle \otimes_R k \cong \mathbb{T}_m$$

and

$$K\langle x \rangle \otimes_K K\langle y \rangle \cong K\langle x, y \rangle$$

hold? It turns out the first one does with  $K = R[\frac{1}{\varpi}]$ . However, the second one doesn't hold. But we have

$$K\langle x \rangle \otimes_K K\langle y \rangle$$

is dense in  $K\langle x, y \rangle$ .

Suppose  $K = \overline{K}$  from now on. Here  $\overline{B}(0, 1) \subset K^n$  is endowed with maximum norm. We denote,

$$K\langle x \rangle$$

as the set of analytic functions on  $\overline{B}(0, 1)$ . What about  $\overline{B}(0, r)$  for  $0 < r < 1$ ? By noting that  $x \leq r \Leftrightarrow \frac{x}{r} \leq 1$ , we have,

$$K\left\langle \frac{x}{r} \right\rangle = \left\{ \sum a_i x^i \mid |a_i r^i| \rightarrow 0 \right\}.$$

is such set of analytic functions. The analytic functions on  $B(0, 1)$  is defined,

$$\bigcap_{0 < r < 1} K\left\langle \frac{x}{r} \right\rangle = \left\{ \sum a_i x^i \mid |a_i r^i| \rightarrow 0, \forall 0 < r < 1 \right\}.$$

**Proposition 2.1** (Universal Property of Tate algebra). *For any complete  $K$ -algebra  $A$  with  $(a_1, \dots, a_n)$  ordered set of  $n$  pb elements in  $A$ , there is a unique continuous morphism of  $K$ -algebras,*

$$f : \mathbb{T}_n \rightarrow A$$

*such that  $f(x_i) = a_i$ .*

**Proposition 2.2.** *Every filter is contained in a maximal filter.*

*Proof.* Consider a sequence of filters,

$$\mathcal{U}_0 \subseteq \dots$$

Then we claim,

$$\mathcal{U} = \bigcup_{i \in I} \mathcal{U}_i$$

is again a filter. Indeed if  $M, N \in \mathcal{U}$ , we have,

$$M \cap N \in \mathcal{U}_l \subseteq \mathcal{U}.$$

Also  $\emptyset \notin \mathcal{U}$ . If  $M \in \mathcal{U}$  and  $M \subseteq N$  then take  $M \in \mathcal{U}$  then  $N \in \mathcal{U}_l \subseteq \mathcal{U}$ . By Zorn's lemma, every filter is contained in an ultrafilter.  $\square$

**Proposition 2.3.** *The direct image of an ultrafilter is an ultrafilter. If  $\iota : A \hookrightarrow X$  then  $\iota_*$  is a bijection from ultrafilters on  $A$  to ultrafilters on  $X$  containing  $A$ .*

*Proof.* Recall that the following statements are equivalent.

1.  $\mathcal{U}$  is an ultrafilter on  $X$
2. For any subset  $M \subseteq X$ , either  $M \in \mathcal{U}$  or  $X \setminus M \in \mathcal{U}$ .
3. If  $M \cup N \in \mathcal{U}$  then either  $M \in \mathcal{U}$  or  $N \in \mathcal{U}$ .

Suppose  $f : X \rightarrow Y$  and  $\mathcal{U}$  is an ultrafilter on  $X$ , then define,

$$f_*\mathcal{U} = \{V \subseteq Y \mid f^{-1}(V) \in \mathcal{U}\}.$$

Easy to verify this is again an ultrafilter. For  $M \subseteq Y$ , note that

$$f^{-1}(M) \sqcup f^{-1}(Y \setminus M)$$

thus exactly one of  $f^{-1}(M)$  and  $f^{-1}(Y \setminus M)$  is in  $\mathcal{U}$  hence one of  $M$  and  $Y \setminus M$  is in  $f_*\mathcal{U}$ .  $\square$

### 3 Week 2

**Lemma 3.1.** *Let  $\mathcal{B}$  be an open basis of the space  $X$ . If  $X$  is  $T_0$  then  $X$  is spectral if and only if every ultrafilter  $\mathfrak{U}$  has a generic limit  $g$  such that  $g \in \Omega \Leftrightarrow \Omega \in \mathfrak{U}$ . If this holds then for any  $\mathcal{B} \ni \Omega$  is quasi-compact.*

#### Problem 5

We already have seen  $\text{Spec } A$  is spectral. It is  $T_0$  and for any ultrafilter  $\mathfrak{U}$  on  $\text{Spec } A$ , define,

$$\mathfrak{g}_{\mathfrak{U}} = \{x \in A \mid \{\mathfrak{p} \in \text{Spec } A \mid x \in \mathfrak{p}\} \in \mathfrak{U}\}$$

is an ideal. This is quite obvious as  $\mathfrak{p}$  is an ideal. That is

$$\{\mathfrak{p} \mid ax \in \mathfrak{p}\} \supseteq \{\mathfrak{p} \mid x \in \mathfrak{p}\}.$$

#### Problem 6

First prove that  $X = \text{Sper } A$  is  $T_0$ . If  $P \neq Q$  then there is  $a \in P \setminus Q$ . Thus  $P \notin \mathcal{P}(-a)$  but  $Q \in \mathcal{P}(-a)$ . Let  $\mathfrak{U}$  be an ultrafilter on  $X$ . We then define,

$$\mathfrak{P} = \{a \in A \mid \{C \in \text{Sper } A \mid a \in C\} \in \mathfrak{U}\}.$$

We would like to verify that  $\mathfrak{P}$  is a generic limit. Let,

$$F(a) := \{\mathfrak{p} \in \text{Sper } A \mid a \in \mathfrak{p}\}.$$

For a square  $b^2$ , we have  $F(b^2) = X$ . Thus  $b^2 \in \mathfrak{P}$ . For  $F(-1) = \emptyset \notin \mathfrak{U}$  thus  $-1 \notin \mathfrak{P}$ .

Let  $a, b \in \mathfrak{P}$  then  $F(a + b) \supseteq F(a) \cap F(b)$ . Thus  $F(a + b) \in \mathfrak{U}$ . Also for  $F(ab)$ . If  $ab \in \mathfrak{P}$  but  $a \notin \mathfrak{P}$ , then

$$\{\mathfrak{p} \mid -b \in \mathfrak{p}\} \supseteq \{\mathfrak{p} \mid ab \in \mathfrak{p}, a \notin \mathfrak{p}\} = F(ab) \cap (X \setminus F(a)).$$

Since  $F(a)$  is not in  $\mathfrak{U}$  thus  $X \setminus F(a)$  is in  $\mathfrak{U}$ . Thus  $F(-b) \in \mathfrak{U}$ .

For  $\mathcal{P}(a) \in \mathcal{B}$ ,  $\mathfrak{P} \in \mathcal{P}(a)$  if and only if  $-a \notin \mathfrak{P}$ . Thus  $F(-a) \notin \mathfrak{U}$ . By definition  $F(-a) \cup \mathcal{P}(a) = X$ . Thus  $\mathcal{P}(a) \in \mathfrak{U}$ . Also  $\mathfrak{P} \in \mathcal{P}(a_1, \dots, a_n)$  then  $\mathfrak{P} \in \mathcal{P}(a_i)$  for all  $i$ . Thus  $\mathcal{P}(a_i) \in \mathfrak{U}$ . In particular  $\mathcal{P}(a_1, \dots, a_n) \in \mathfrak{U}$ . By the lemma  $X$  is spectral and all elements in  $\mathcal{B}$  are quasi-compact.

#### Problem 7

Clearly  $X$  is  $T_0$  since for any  $x \neq y \in X$ . Without loss of generality, we assume  $x \neq \eta$ . Then  $X \setminus \{x\}$  does the work. Let  $\mathfrak{U}$  be an ultrafilter on  $X$ . If  $U \in \mathfrak{U}$ , then  $X \setminus U$  is finite thus there is  $\{x_i\} \in \mathfrak{U}$ . If  $U$  is cofinite, then  $x_i \in U \Leftrightarrow U \in \mathfrak{U}$ . So  $U$  is quasi-compact. If  $\mathfrak{U}$  contains all non-empty open sets then  $\eta$  the generic limit. So  $X$  is spectral and cofinite subsets are quasi-compact.

#### Problem 8

Note that if  $x \neq \eta$ , then  $\{x\}$  is open thus this is an open neighborhood. On the other hand, the neighbourhood basis of  $\eta$  is the basis open basis is the neighborhood basis as  $X \setminus U_i \ni \eta$  then  $U_i = \emptyset$  thus it is the whole.

**Theorem 3.1.** *If  $X$  is quasi-compact and Hausdorff and denote  $C(X)$  be the set of continuous functions  $X \rightarrow \mathbb{R}$ . Then  $C(X)$  is a commutative unital ring. Then there is a one-to-one correspondence between  $X$  and the maximal ideals of  $C(X)$ , mapping  $x \mapsto \mathfrak{m}_x = \{f \mid f(x) = 0\}$ .*

Are there non-maximal prime ideal? The answer is if  $X$  is first countable, then non-maximal prime ideal exists if and only if  $X$  is an infinite set.